Melbourne Airbnb Market Screening

CMCE30005 Final Project Report | Group 2

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Project repository: https://github.com/maksym-xu/CMCE30005-TheNextChapter/tree/main

1. Introduction

Our client plans to lease Melbourne homes and let them on Airbnb as short stays (rent-to-rent). Rent and setup costs come before any guest income, so a poor lease ties up scarce funds. Our main answers are:

- **Where:** City of Melbourne two- and three-bedroom apartments, then Yarra Ranges three-bedroom houses. These groups had the highest shares of busy homes (30 or more reviews a year, about 180 booked nights).
- **Which homes:** those that can take one-night stays at a price at or below similar listings. Half of such homes were busy, against a third of all homes in these groups.
- **Which leases:** only those whose weekly costs sit below the group's typical weekly revenue (AUD 590–930).

2. Problem Definition and Objectives

The business question is: **where should a rent-to-rent operator with limited funds search, and which homes should it inspect first?** We set three objectives:

1. Rank property groups by their share of busy homes, and express this in nights and revenue.
2. Find which home settings go with being busy, and whether they help choose homes within the leading groups.
3. Turn the results into a search order, a checklist and a four-week inspection plan.

A **group** combines council area, dwelling type and bedroom count. A home is **busy** if it received at least 30 guest reviews in the year before data collection. Inside Airbnb (n.d.) assumes half of stays are reviewed, so with three-night stays, 30 reviews means about 180 booked nights, roughly half the year.

Why reviews? Empty nights are the operator's main risk. Review counts are an indirect record of past guest activity and a public indicator of demand. Turning them into booked nights needs assumptions about review rates and stay length, and shorter stays bring more reviews for the same nights. Review counts show neither profit (rents and costs are not in the data) nor satisfaction, so we translate them into nights and revenue and check guest ratings and review text (Section 5.1).

Since the interim report (in the repository), we replaced its random forest with one explainable logistic regression and moved technical checks to Appendix A.

3. Data Description and Preparation

We use one Inside Airbnb snapshot of Melbourne, supplied through the subject LMS and collected 17 June–1 July 2026 (Inside Airbnb, 2026). It holds 25,728 listings and 1,026,690 dated reviews. We did not use the calendar, because it cannot separate bookings from nights the host blocked.

Cleaning converted prices to numbers, standardized missing entries, counted amenities and rebuilt each home's annual review count from review dates. Price is missing for 6,553 listings (25%) and bedrooms for 4,679 (18%); 4,477 listings (17%) have never been reviewed.

We kept homes a small operator could lease (Table 1):

- entire apartments, houses or townhouses with one to three bedrooms;

- prices of AUD 30–1,500, since unpriced and extreme listings are mostly inactive: 77.8% of those removed had no review in the past year, against 17.0% of those kept;
- a first review at least a year before collection, so each home has a full year of reviews;
- groups of at least 50 homes, so each group's share rests on enough homes.

To avoid setting the target with the same homes we compare, we first set aside one host in five (1,342 homes). Within our 14 groups, the busiest quarter of these set-aside homes had 30 or more reviews, so 30 became the target. The final sample has 3,873 homes, 1,726 hosts and 14 groups in six council areas.

Table 1. From supplied listings to analysis sample

Step	Homes
All listings	25,728
Entire homes with one to three bedrooms	16,576
Apartments, condos, houses and townhouses	14,895
Priced AUD 30–1,500	11,099
First review at least a year before collection	6,891
After setting aside 1,342 reference-host homes	5,549
Analysis sample: 14 groups with at least 50 homes	**3,873**

4. Methodology

Descriptive (Objective 1). A simple summary, not a model, ranks the 14 groups by their share of busy homes, adding Inside Airbnb's estimated nights and revenue.

Predictive (Objective 2). A logistic regression estimates each home's chance of being busy from six inputs, holding the others fixed (Table 2; R Core Team, 2026). This separates location from how a home is run. We also test a simpler two-setting rule: one-night stays, and a price at or below the group median.

Table 2. Outcome and inputs of the model

Variable	Definition
Outcome	1 if the home had at least 30 reviews in the year, otherwise 0
Council area	One of six
Dwelling type	Apartment, or house/townhouse
Bedrooms	One, two or three
Price compared with similar homes	Price relative to the group median, per 10%
Accepts one-night stays	Minimum stay of one night (yes or no)
Amenity count	Amenities listed, per ten

Prescriptive (Objective 3). We turn the results into a checklist, a cost test and an inspection plan.

Checks against luck. Three checks each answer one question:

1. *Is the group ranking stable?* We redraw hosts at random 1,000 times.
2. *Does the model work on unseen homes?* In 50 tests, it is built on 70% of hosts and picks a quarter of each priority group's homes among the other 30%, against a reduced model (no price or minimum stay) and random picking.
3. *Do leading groups stay ahead?* We rank groups on one year and follow them into the next.

Homes of one host stay together in every check, because they can perform alike. The model assumes each 10% price step, or ten more amenities, changes the odds of being busy by the same proportion at any level; Appendix A finds this roughly holds and the inputs barely overlap.

5. Analysis and Results

5.1 Data exploration: activity is uneven, and busy homes are better rated

Guest activity is very uneven (Figure 1). The median home received 15 reviews in the year; a quarter (981 homes, 25.3%) were busy, and 334 (8.6%) received none.

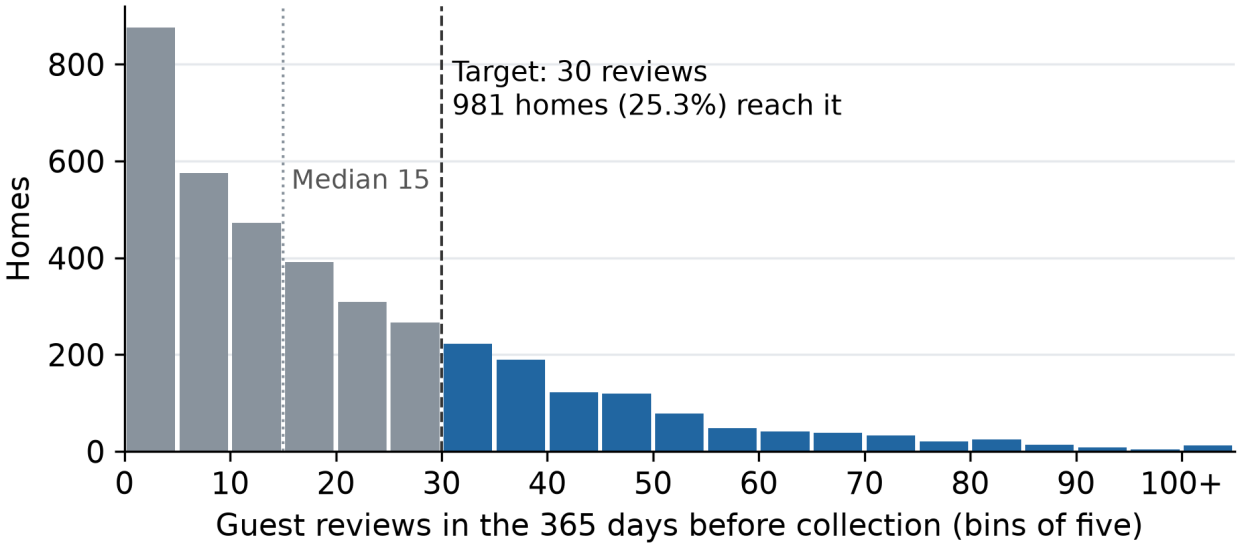


Figure 1. Annual reviews per home, 3,873 analysis homes. Source of all figures and tables: Inside Airbnb (2026); Group 2 analysis.

Busy homes were slightly cheaper, but far more often accepted one-night stays and sat in the City of Melbourne (Table 3). Their much higher revenue estimate reflects their review counts, since revenue is estimated from reviews.

Table 3. Busy homes compared with the rest

Characteristic	Busy (30+ reviews)	Not busy	All homes
Homes	981	2,892	3,873
Median nightly price (AUD)	254	266	262
Accept one-night stays	67.7%	37.5%	45.1%
In the City of Melbourne	76.9%	59.6%	64.0%
Median estimated nights (cap 255)	255	60	96
Median estimated revenue (AUD a year)	58,400	16,600	24,700
Mean lifetime guest rating (out of 5 stars)	4.84	4.69	4.73
Reviews in the year with a negative tone	1.6%	2.9%	2.2%

Busy homes also had a higher mean lifetime rating (4.84 against 4.69 stars), and a smaller share of their reviews was classified as negative (1.6% against 2.9%). We scored all 78,288 reviews from that year with VADER, a word-list tool for English text (Hutto & Gilbert, 2014); about 90% were classified as positive in both groups. The text score is a supporting indicator, not a tested measure of satisfaction.

5.2 Descriptive analysis: City apartments lead

Shares of busy homes range from 8.2% to 36.6% across groups, and three groups lead (Figure 2, Table 4). The City groups stayed in the top three in 94% and 90% of host redraws, but Yarra Ranges houses in only 64%, so they are a less certain option (more checks in Table A3).

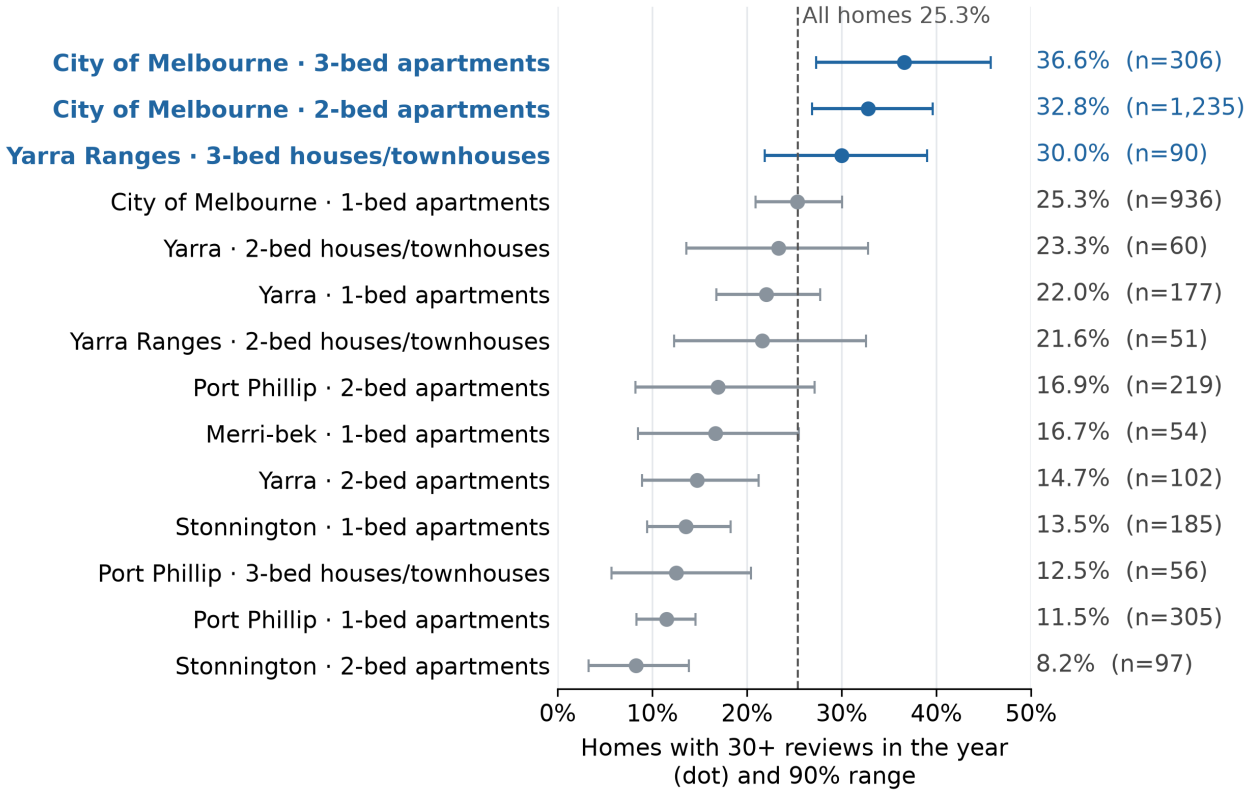


Figure 2. Share of busy homes in all 14 groups, with 90% ranges from redrawing hosts; priority groups in blue.

Table 4. Priority groups: median estimated nights and annual revenue (AUD)

Group	Homes	Share busy	Nights, all homes	Revenue, all homes	Revenue rank	Revenue, not busy
City 3-bed apartments	306	36.6%	126	48,600	1	32,100
City 2-bed apartments	1,235	32.8%	108	30,600	5	16,600
Yarra Ranges 3-bed houses	90	30.0%	120	46,100	2	31,100
All 14 groups	3,873	25.3%	96	24,700	–	16,600

Activity and revenue rank groups differently. On median revenue, Yarra Ranges houses rank second of 14 and City two-bedroom apartments fifth, because the houses charge more a night (AUD 426 against 285). Without rents, revenue cannot show which lease is better, so activity guides the search and revenue tests each lease.

5.3 Predictive analysis: one-night stays and price go with busy homes

Holding other inputs fixed, a typical City two-bedroom apartment has a **42%** chance of being busy with one-night stays, but **18%** with a two-night minimum or longer (Figure 3, Table A1). Pricing 20% above similar homes lowers the chance to **36%**, and ten more amenities raise it to **48%**. The same apartment would score 26–27% in Port Phillip or Stonnington. These are associations, not guarantees.

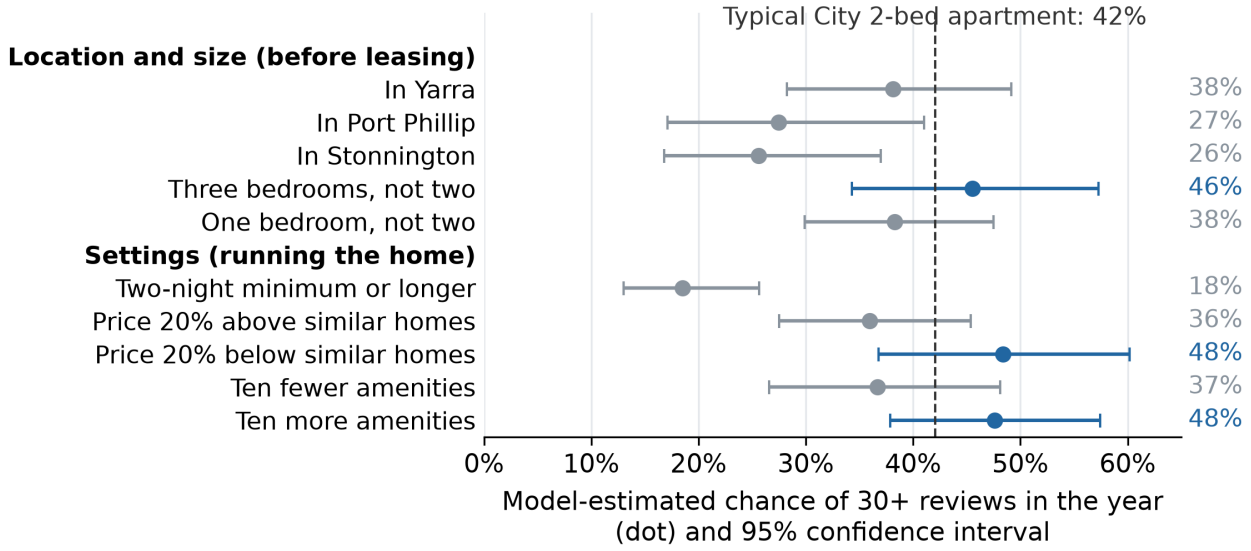


Figure 3. Chance that a typical City two-bedroom apartment (one-night stays, median price and amenities) is busy as one input changes; 95% confidence intervals.

A two-setting rule turns this into a simple check (Table 5). In the priority groups, half (50.4%) of homes with one-night stays priced at or below their group median were busy, against a third (33.4%) of all their homes and 9.3% with neither setting. Only 20 of 90 Yarra Ranges houses accept one-night stays, so the rule rests mainly on City apartments.

Table 5. Share of busy homes in the three priority groups, by two settings

Minimum stay	Price at or below group median	Price above group median
One night accepted	50.4% (462 homes)	36.1% (477 homes)
Two nights or more	30.2% (358 homes)	9.3% (334 homes)

The rule picks 28% of homes in this sample and suits a quick first check. The model adds evidence from homes it had not seen: in 50 tests, 49.6% of its picks were busy, against 38.6% for the reduced model and 32.4% for random picking (Figure 4, Table A3; Table A2 covers all 14 groups). The model's main value is in finding which settings matter.

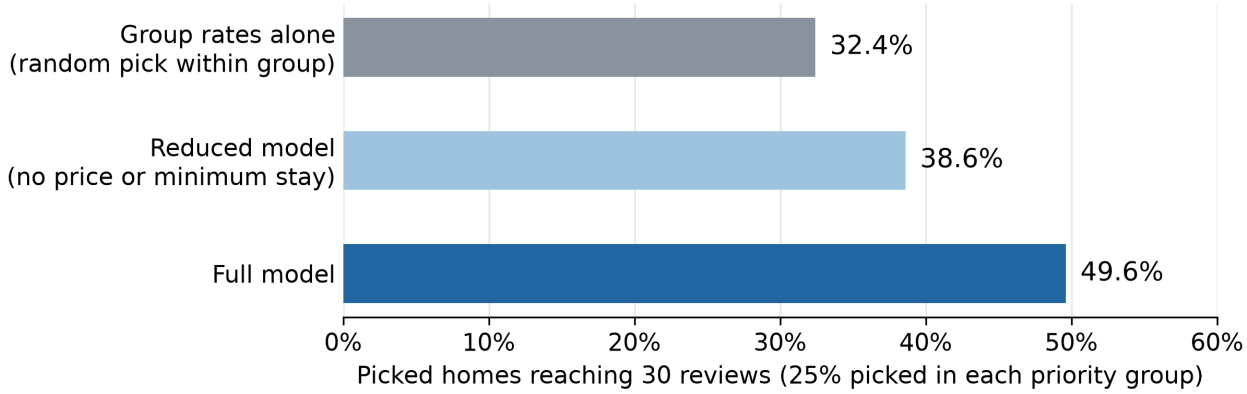


Figure 4. Share of picked homes that were busy (a quarter of each priority group), averaged over 50 tests.

5.4 Over time: the leading groups stayed ahead

We repeated the ranking on the earlier year (ten groups, 2,657 homes with two years of reviews), with a 34-review bar set the same way. The same three groups led: 33.6% of their homes reached it, against 19.9% elsewhere (Figure 5). On the same bar a year later, they still led (27.1% against 14.9%), although Yarra Ranges houses fell to 21.4%.

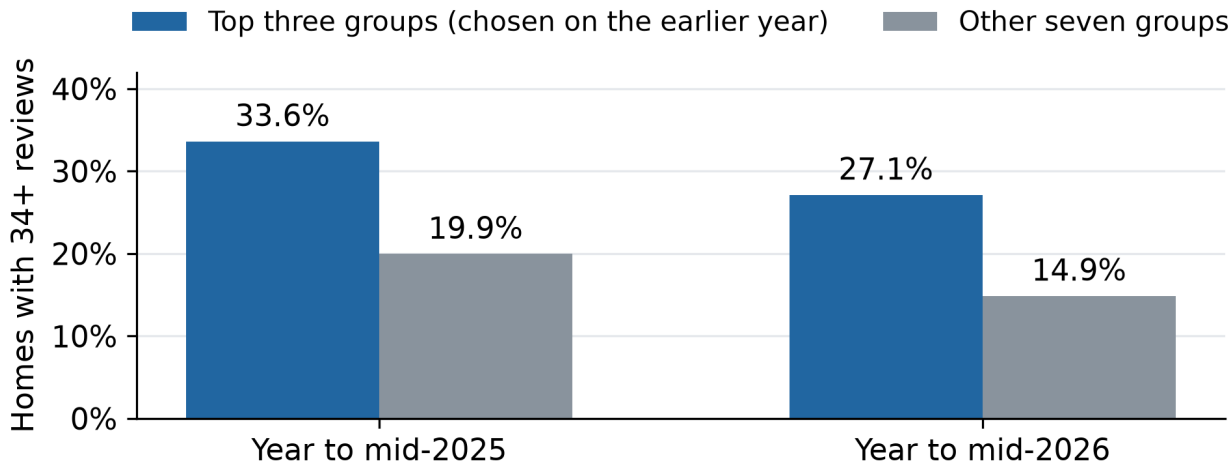


Figure 5. The same 2,657 homes in two consecutive years; target 34 reviews.

6. Key Findings and Recommendations

Key findings

1. **Where:** City of Melbourne two- and three-bedroom apartments had the highest shares of busy homes, then, less certainly, Yarra Ranges three-bedroom houses. All three stayed ahead a year later and had above-median revenue (Figures 2 and 5, Table 4).
2. **What goes with being busy:** one-night stays and a price at or below similar homes (Figure 3, Table 5), though shorter stays partly inflate review counts.
3. **Guest feedback:** busy homes had a higher mean rating and a smaller share of reviews classified as negative (Table 3).
4. **Which home first:** the two settings give a quick first check (Table 5). Tested on unseen hosts, 49.6% of the model's picks were busy, against 32.4% for random picking (Figure 4).

Recommendations

1. **Search City of Melbourne two- and three-bedroom apartments first, then Yarra Ranges three-bedroom houses** (Figure 2, Table 4). Between leases that pass Recommendation 3, choose the larger cash-flow margin; if margins are similar, choose the group with the higher share of busy homes.
2. **Favour homes that can take one-night stays at a price at or below similar listings** (Figure 3, Table 5); the evidence is strongest for City apartments. At inspection, check that building rules allow one-night stays and that this price covers costs. Trial these settings and track the results (Table 6).
3. **Drop leases whose weekly costs exceed the group's typical revenue** (Figure 6). Typical estimated weekly revenue is **AUD 590** for City two-bedroom apartments, **930** for City three-bedroom apartments and **890** for Yarra Ranges houses, but only half of homes reach it; for homes that were not busy it was **AUD 320, 620 and 600**. Assess each remaining lease with its own cash-flow plan, and hold cash to cover a year's shortfall at that lower revenue.

Table 6. Four-week inspection plan

Timing and owner	Action	Decision condition
Week 1 (Operator)	Pool ten candidate homes in the priority groups; check one-night stays and price	Inspect homes meeting both first, in group order
Weeks 2–3 (Operator & Analyst)	Get written landlord consent; check owners' corporation rules on stays under 28 days (Consumer Affairs Victoria, n.d.); negotiate rent; get cost quotes	Replace homes that fail any check
Week 4 (Client)	Cash-flow plan: quotes and Victoria's 7.5% short-stay levy (State Revenue Office Victoria, n.d.) against typical and not-busy revenue (Recommendation 3)	Sign only if the plan covers setup costs, required return and a year at the not-busy level
First 30 days (Operator)	Record paid nights, stay length, workload, net cash flow and reviews	Expand only if results match the plan

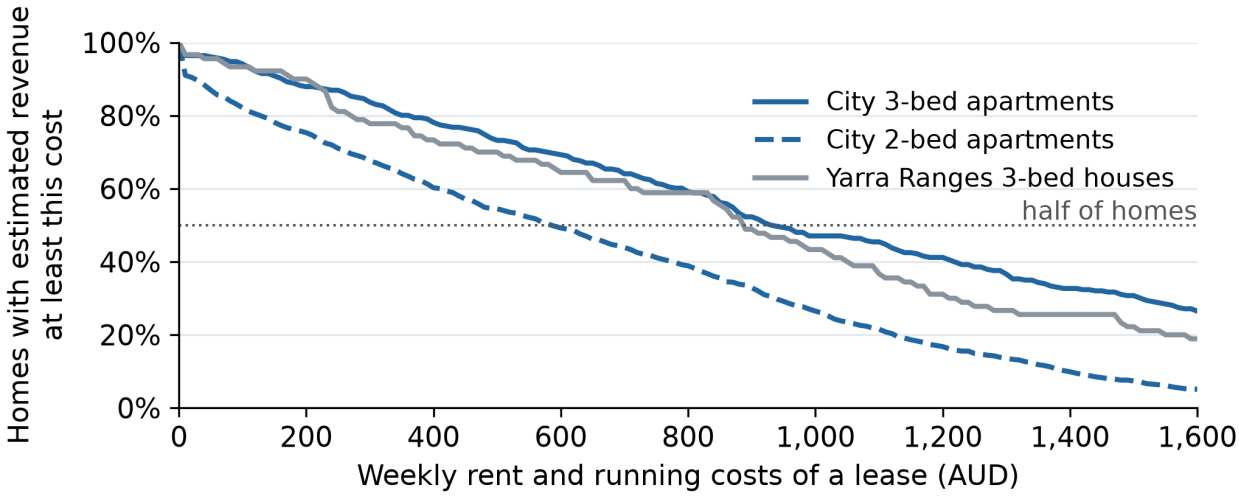


Figure 6. Share of each priority group's homes whose estimated weekly revenue reached a given weekly cost.

Worked example (illustrative costs). A City two-bedroom apartment costing **AUD 620** a week exceeds its group's typical AUD 590, so it is dropped. A Yarra Ranges house costing **AUD 700** clears its typical AUD 890. Earning like a home that is not busy (AUD 600), it would need a buffer of AUD 5,200 for the year.

7. Limitations and Future Work

- **Reviews measure activity indirectly.** Inside Airbnb's 50% review rate sits within published estimates of 30.5%–72% (Inside Airbnb, n.d.), so nights and dollar figures are rough. Ratings are lifetime averages, and VADER does not translate reviews written in other languages.
- **One snapshot of surviving homes.** Departed, new, unpriced and outer-area listings are missing, and revenue uses June–July prices. The two-year check uses the same snapshot.
- **Short stays inflate reviews.** A one-night minimum allows more stays for the same nights booked.
- **Associations, not causes.** Settings may respond to past performance, and we chose the inputs, rule and priority groups after seeing all homes.
- **No rent, cost or regulation data.** Recommendation 3 relies on the client's own quotes.

Future work should:

1. collect rent and cost quotes and track paid nights and net cash flow, to test whether busy homes are also profitable;
2. retest the groups, model and two-setting rule, unchanged, on the next snapshot;
3. read the negative reviews of priority-group homes to find problems an operator can fix.

8. References

Consumer Affairs Victoria. (n.d.). *Making rules to ban short stay accommodation*. Retrieved October 16, 2026, from https://www.consumer.vic.gov.au/housing/owners-corporations/rules/making-rules-to-ban-short-stay-accommodation

Huang, E., & Xu, M. (2026). *Melbourne Airbnb market screening* [Source code]. GitHub. https://github.com/maksym-xu/CMCE30005-TheNextChapter/tree/main

Hutto, C., & Gilbert, E. (2014). VADER: A parsimonious rule-based model for sentiment analysis of social media text. *Proceedings of the International AAAI Conference on Web and Social Media, 8*(1), 216–225. https://doi.org/10.1609/icwsm.v8i1.14550

Inside Airbnb. (n.d.). *Data assumptions*. Retrieved October 16, 2026, from https://insideairbnb.com/data-assumptions/

Inside Airbnb. (2026). *Melbourne, Victoria, Australia: Listings and reviews* (June 2026 snapshot, via CMCE30005 LMS) [Data set]. https://insideairbnb.com/get-the-data/

R Core Team. (2026). *R: A language and environment for statistical computing* (Version 4.5.3) [Computer software]. R Foundation for Statistical Computing. https://www.R-project.org/

State Revenue Office Victoria. (n.d.). *Understanding the short stay levy*. Retrieved October 16, 2026, from https://www.sro.vic.gov.au/owning-property/short-stay-levy/understanding-short-stay-levy

AI acknowledgement

ChatGPT, Gemini and Claude Code assisted with writing and coding; Eric Huang and Maksym Xu reviewed the outputs.

Author contributions

Maksym Xu led data preparation, modelling and validation; Eric Huang led report drafting and review. Earlier team contributions are recorded in the repository guide.

Appendix A. Model Details and Further Checks

Group median prices, and the most common minimum stay used to fill gaps, come only from the homes each model was built on. In Table A1, each odds ratio multiplies the odds of being busy (busy homes per home that is not busy), averaged over the 50 fits; above 1 raises the chance, below 1 lowers it.

Table A1. Odds ratios across 50 fits

Input (compared with)	Odds ratio	Range across fits
Accepts one-night stays (no)	3.27	2.53–4.05
Ten more amenities	1.27	1.15–1.41
Price 10% above similar homes	0.88	0.83–0.94
Council area (City of Melbourne)		
Port Phillip	0.50	0.33–0.66
Stonnington	0.48	0.26–0.68
Yarra	0.83	0.59–1.11
Yarra Ranges	1.22	0.76–1.97
Merri-bek	0.75	0.36–1.19
House or townhouse (apartment)	0.78	0.57–1.30
Two bedrooms (one)	1.16	0.89–1.41
Three bedrooms (one)	1.31	0.86–1.84

Ranges show stability across fits, not confidence intervals; those including 1 (for example, Yarra Ranges) leave the direction uncertain. Across the 50 tests on all 14 groups, the model ranked a random busy home above a random home that was not busy 71% of the time, against 60% for group rates (AUC 0.71 and 0.60). Treating same-host homes as related widens margins of error to 1.1–2.6 times ordinary ones. Inputs overlap little (variance inflation factors below 3). An ordinary test of curved price and amenity effects suggests slight curvature ($p = 0.03$; amenities alone, $p = 0.011$), but this is unclear once same-host homes are treated as related (amenities, $p = 0.17$). Full evaluation tables are in the repository (Huang & Xu, 2026).

Table A2. Hits and misses: top quarter picked across all 14 groups

Selection	Busy	Not busy	Total
Picked by model	423	546	969
Not picked	558	2,346	2,904

Each home was scored by a model built without its host (five folds of hosts). Of the 969 picked, 43.7% were busy (precision), against 25.3% overall, and they included 43.1% of busy homes (recall). Accuracy (71.5%) is not a useful guide here, because calling every home "not busy" scores 74.7%.

Table A3. Further checks

Check	Result
Top three in 1,000 host redraws	City 3-bed 94%, City 2-bed 90%, Yarra Ranges 64% of redraws
Without the price rule, or without the one-year review rule	Same three groups; Yarra Ranges houses rank second
Minimum group size of 30 or 75 homes	Same three groups, same order
Two-year check with hosts weighted equally	Later-year lead of 8.7 points
Model against random picking, 50 tests	Model better in 46 tests
Model picks 10%, 25% or 50% of each priority group	54.5%, 49.6% and 43.1% of picks busy; they include 17.1%, 38.3% and 66.7% of busy homes
Reduced model against random picking, by group	City 2-bed 38.9% vs 31.7%; City 3-bed 33.9% vs 35.9%; Yarra Ranges 49.7% vs 30.4%